Long-term walnut supplementation without dietary advice induces favorable serum lipid changes in free-living individuals.

BACKGROUND/OBJECTIVES: Walnuts have been shown to reduce serum lipids in short-term well-controlled feeding trials. Little information exists on the effect and sustainability of walnut consumption for longer duration in a free-living situation. SUBJECTS/METHODS: A randomized crossover design in which 87 subjects with normal to moderate high plasma total cholesterol were initially assigned to a walnut-supplemented diet or habitual (control) diet for a 6-month period, then switched to the alternate dietary intervention for a second 6-month period. Each subject attended seven clinics 2 months apart. At each clinic, body weight was measured, and in five clinics (months 0, 4, 6, 10 and 12), a blood sample was collected. RESULTS: Our study showed that supplementing a habitual diet with walnuts (12% of total daily energy intake equivalent) improves the plasma lipid profile. This beneficial effect was more significant in subjects with high plasma total cholesterol at baseline. Significant changes in serum concentrations of total cholesterol (P=0.02) and triglycerides (P=0.03) were seen and nearly significant changes in low-density lipoprotein cholesterol (LDL-C) (P=0.06) were found. No significant change was detected in either high-density lipoprotein (HDL) cholesterol LDL to HDL ratio. CONCLUSIONS: Including walnuts as part of a habitual diet favorably altered the plasma lipid profile. The lipid-lowering effects of walnuts were more evident among subjects with higher lipid baseline values, precisely those people with greater need of reducing plasma total and LDL-C.